Habtamu Asefa

Software Engineer | Full Stack, Mobile and AI Agents

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GitHub: github.com/habasefa | Portfolio | LinkedIn | Open to remote work and relocation

Software engineer with four years of experience building mobile applications, backend services and AI agents. Built PrepX's React Native app and tutoring agent, contributed to its Next.js dashboard and FastAPI backend, and led a four-person engineering team.

Technical Skills

Languages: TypeScript, JavaScript, Python, SQL

Web and mobile: React, Next.js, Tailwind CSS, React Native, Expo

Backend and data: Node.js, Express.js, NestJS, FastAPI, PostgreSQL, SQLAlchemy, Prisma, SQLite

AI and automation: Mastra, Agno, LangGraph, n8n, LLM APIs, retrieval-augmented generation, Qdrant

Delivery and real time: Git, Docker, Linux, AWS EC2/S3, GitHub Actions, Socket.IO, WebRTC

Experience

PrepX | Lead Software Engineer | 2024–present

- Built and launched the React Native/Expo mobile app; students answer approximately 30,000 questions per week, with approximately 200 students joining weekly.
- Built a retrieval-augmented tutoring agent using Node.js/NestJS and Mastra, with Qdrant and FastEmbed retrieval over textbooks and past exams and source references in answers.
- Implemented offline content storage with SQLite, streamed AI responses and live competition flows; integrated Typesense search across 250,000+ items, including questions, curriculum topics and schools.
- Contributed to the Next.js/React dashboard and FastAPI backend using SQLAlchemy and PostgreSQL; led a four-person engineering team.
- Implemented an Agno content-generation workflow linking research, notes, quizzes, worksheets and flashcards, with explicit context passing between agents.

Software Engineer | Client Projects | 2022–present

- Dymnd: worked on real-time habit tracking and an AI habit-development advisor.
- Themis Bar: resolved critical mobile bugs, upgraded libraries and delivered feature updates for the bar-exam preparation app.
- Remimb: contributed to a React Native companion application for a Bluetooth keyring.
- Marketplace application: implemented React Native/Expo screens for product browsing, cart, checkout and orders.

Liqawnt Labs | AI Research Engineer | 2026–present

- Fine-tuning Pocket TTS on 200 hours of Amharic speech; built an internal benchmark using word error rate and native-speaker ratings for model evaluation.

Selected Projects

react-native-ajora | Author and Maintainer. Published a React Native AI chat library used in PrepX, supporting AG-UI streaming, tool-call rendering and client-side tool execution. Source

GoNumbers | Telecom Application. Developed React Native calling features using SIP.js, FreeSWITCH, WebRTC and native call controls; worked with an Express.js/Prisma API.

UlcerGuard | Engineering Thesis. Connected Arduino pressure sensors to React Native over Bluetooth and streamed readings through Node.js/Express and Socket.IO to a React clinician dashboard. Source

Education

Addis Ababa University / AAiT | BSc in Biomedical Engineering | 2024